

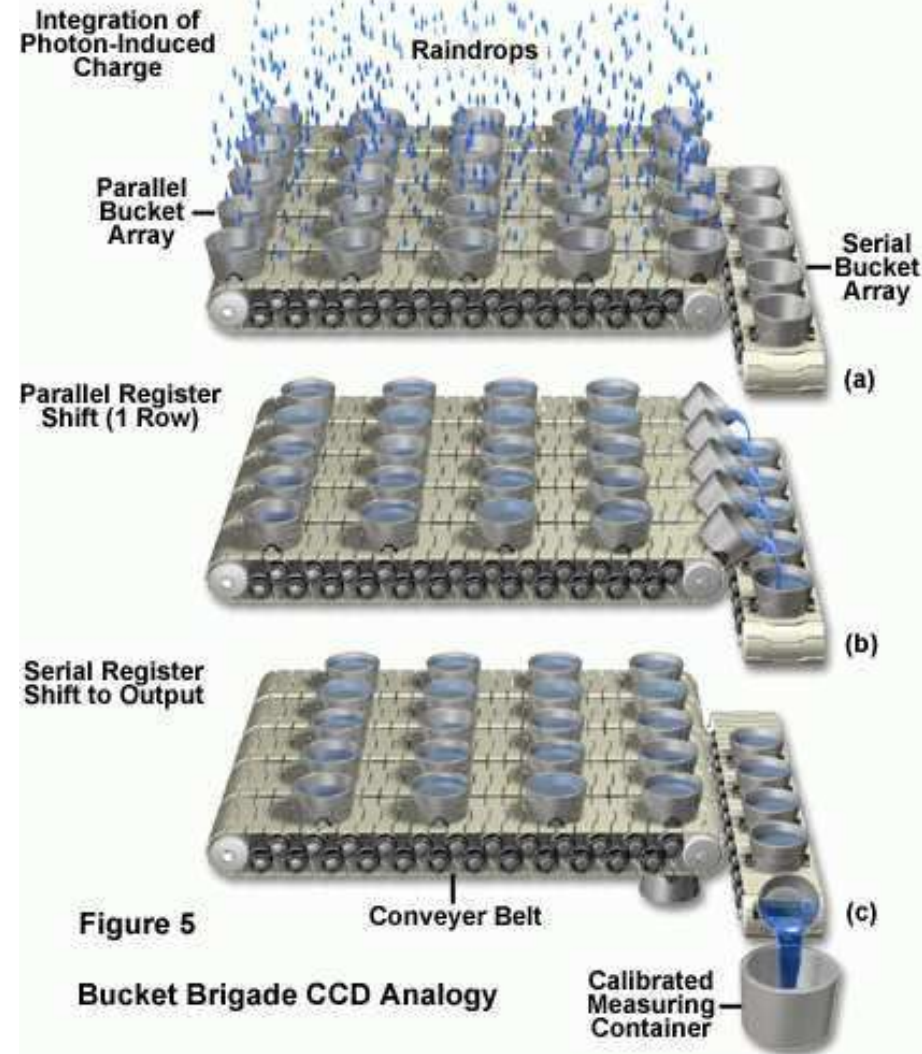
CCD Basics - part 2

gain, readnoise, data reduction & CCD equation

CCDs - quick recap

To measure charge in each pixel:

- 1) Move all rows in parallel; every step one row is moved into a special “serial register”
- 2) Then shift only pixels in serial register towards a special pixel connected to an analog-to-digital converter
- 3) Once all pixels in one row have been digitized, move all rows another step
- 4) Repeat until all rows and pixels have been digitized

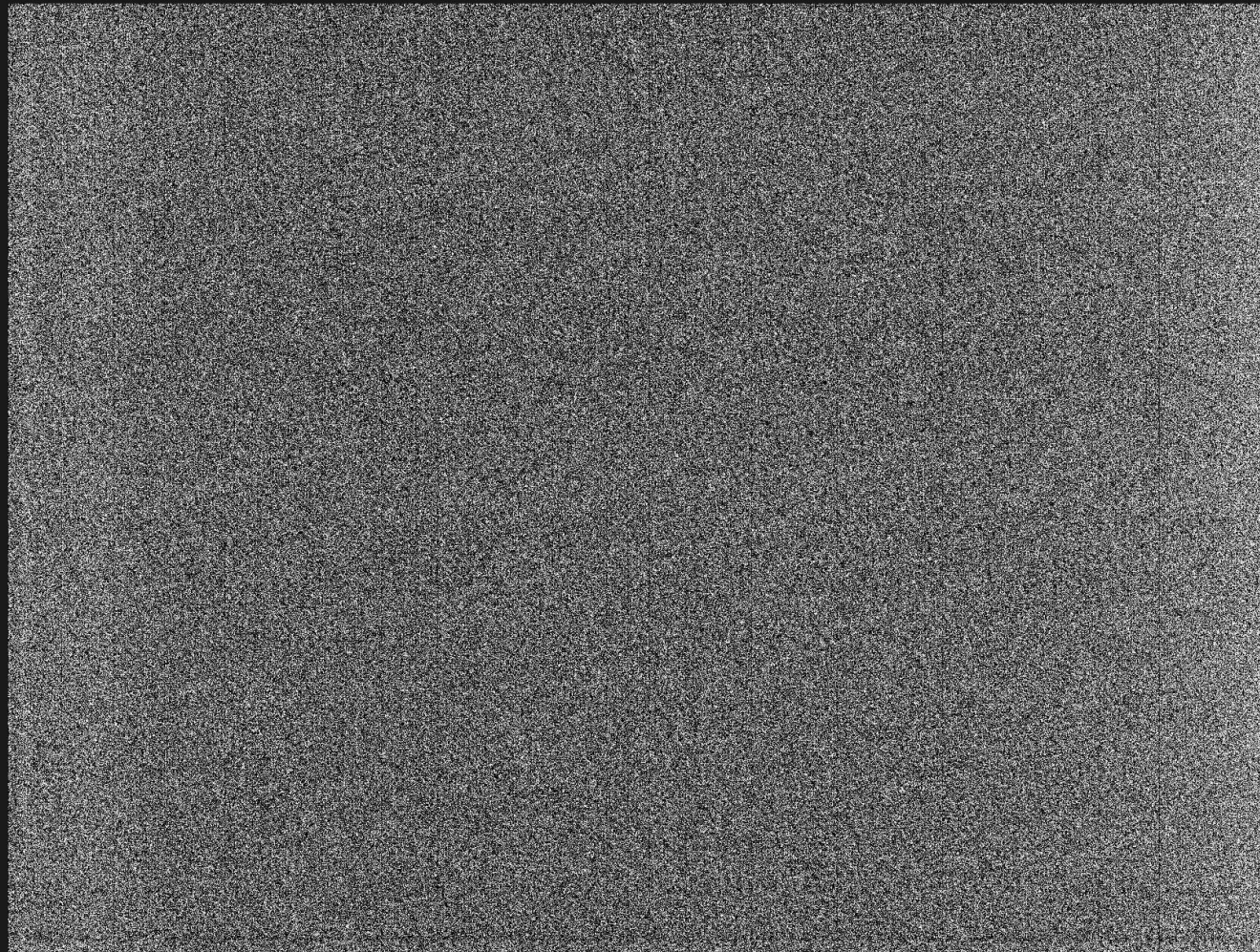


Bias

single frame:

mostly noise
(which is good!)

no discernable
features
(maybe some
elevated levels
near the edges)



Bias frames

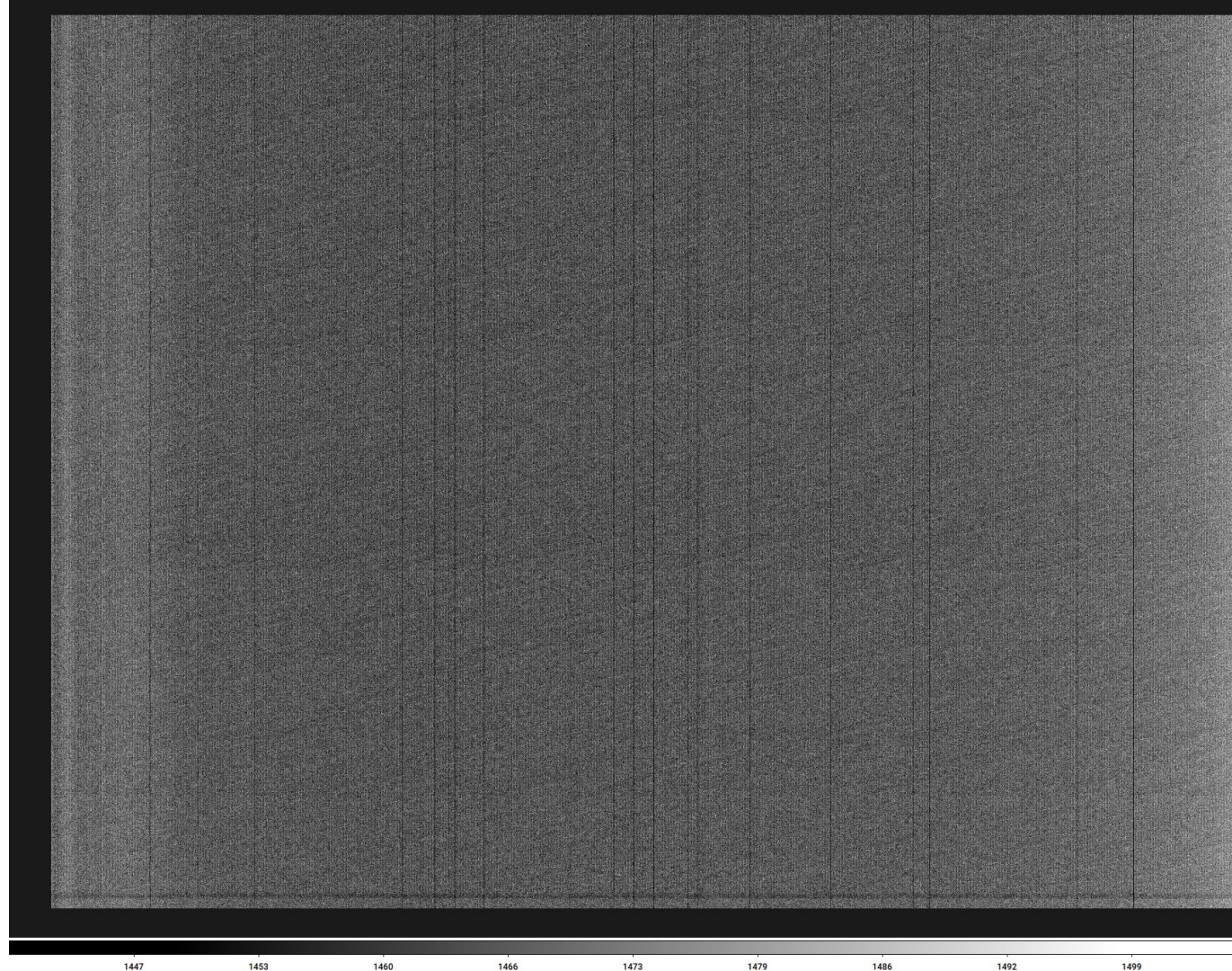
Average of 25
frames:

Elevated levels
near the edges

some banding

several column
defects

horizontal feature
near the bottom



1447

1453

1460

1466

1473

1479

1486

1492

1499

Read-noise and gain

Since CCDs are counting photons, the resulting noise follows a Poissonian distribution:

$$\sigma[e^-] = \sqrt{N[e^-]}$$

However, what we measure from the detector are ADUs, where

$$1 \text{ ADU} = \text{gain} * N[e^-]$$

Therefore, in observational terms:

$$\sigma[\text{ADU}] = \text{gain} * \sqrt{(\text{ADU}/\text{gain})} = \sqrt{\text{ADU}} \times \sqrt{\text{gain}}$$

Read-noise and gain

In practice we also need to consider noise introduced by the readout electronics:
⇒ read noise:

$$\begin{aligned}\sigma^2(\text{total}) &= \sigma^2(\text{read-noise}) + \sigma^2(\text{signal}) \\ &= \sigma^2(\text{read-noise}) + N[e^-]\end{aligned}$$

Estimating read-noise

Read-noise levels (in ADU) can be derived from bias frames that have no signal:

Method 1: Calculate standard deviation across all pixels in a frame

Pro: need only a single frame

Con: Inaccurate if bias has some structure or lots of bad pixels

Method 2: Calculate standard deviation for each pixel across a sequence of bias frames

Pro: Works in all cases

Con: Needs a lot more data

Estimating gain

Earlier we saw: $\sigma^2(\text{total}) = \sigma^2(\text{read-noise}) + \sigma^2(\text{signal})$

In the special case where signal is high enough to neglect the read-noise, we can derive the gain from signal and noise:

$$S_{ADU} = \frac{N[e^-]}{g} \quad \sigma_{ADU} \frac{\sqrt{N[e^-]}}{g} \quad \longrightarrow \quad \sigma_{ADU}^2 = \frac{N[e^-]}{g^2}$$

combining these:

$$\frac{\sigma_{ADU}^2}{S_{ADU}} = \frac{1}{g} \rightarrow g = \frac{S_{ADU}}{\sigma_{ADU}^2}$$

Gain and read-noise (the better way)

In real data, read-noise will never quite be negligible, so we have to take it into account

We can combine bias frames with zero signal with well illuminated data (flat-fields). However, the latter show some large-scale and pixel-to-pixel variations. Using differences between pairs of images eliminates such differences.

Janesick's method (Janesick 2001) allows to estimate gain and readnoise simultaneously, from a pair of biases and flatfields.

Janesick's method

B_1 and B_2 are a pair of bias frames, and F_1 and F_2 are a pair of flat-fields

$$\Delta_B = B_1 - B_2 \qquad \Delta_F = F_1 - F_2$$

$$g = \frac{(\overline{F_1 + F_2}) - (\overline{B_1 + B_2})}{(\sigma_{\Delta_F}^2 - \sigma_{\Delta_B}^2)} \quad [e^- / ADU]$$

$$R = g \times \frac{\sigma_{\Delta_B}}{\sqrt{2}} \quad [e^-]$$

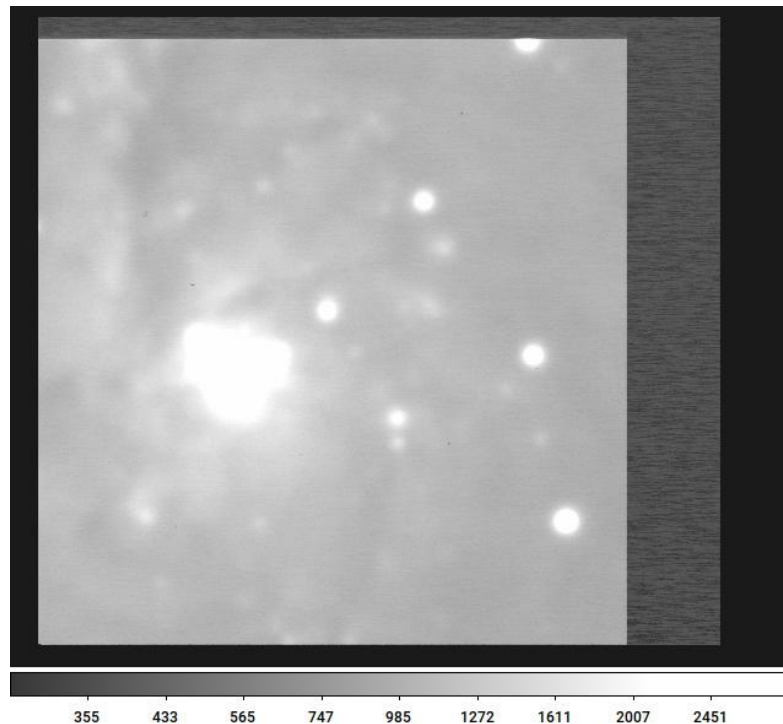
where g and R are gain and read-noise averaged over all pixels in the frame

Bias & Overscan

Analog-to-digital converters prefer not to measure signals very close to zero

- **Add** a constant pedestal voltage/signal before digitization
- Bias image → needs to be **subtracted** before everything else

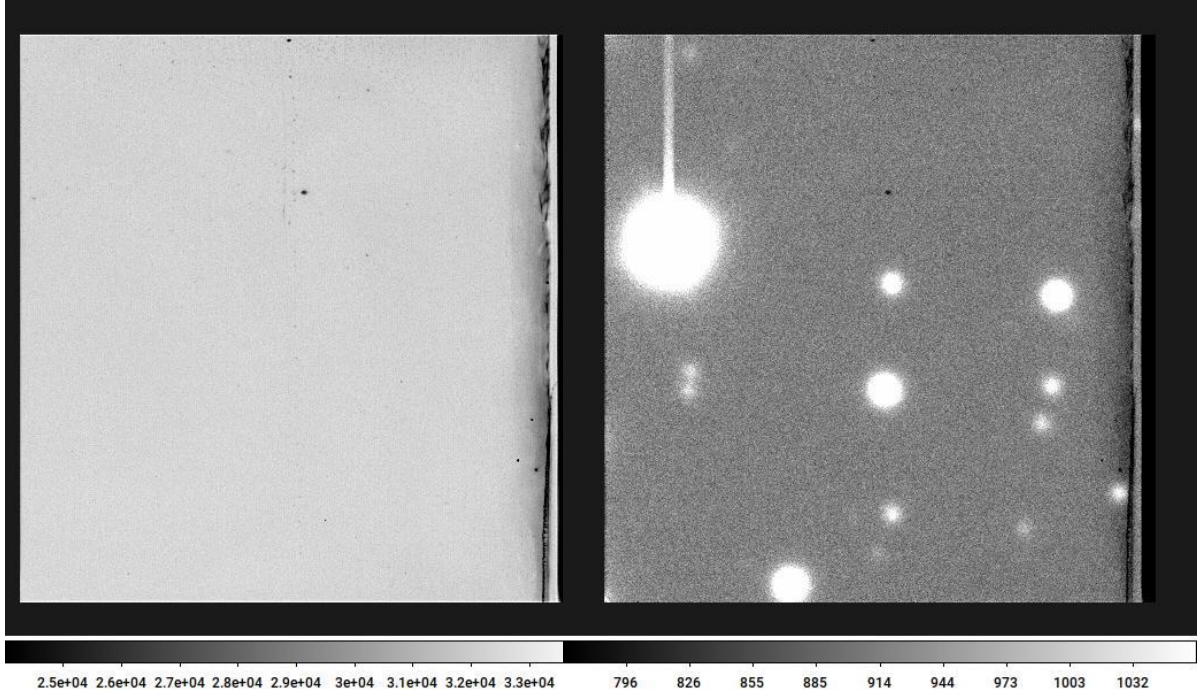
Alternatively: Read out more pixels than have actual signal → Overscan



Flatfielding

Not all pixels are the same

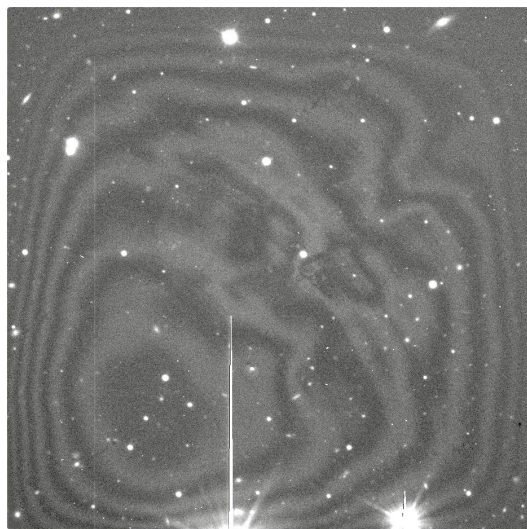
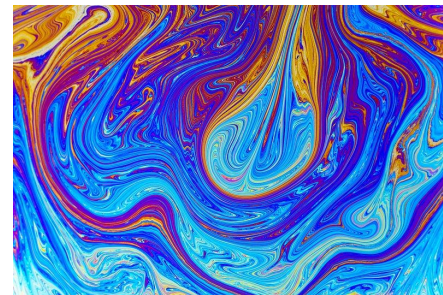
- Take images of white screen or twilight sky
- all pixels should be the same



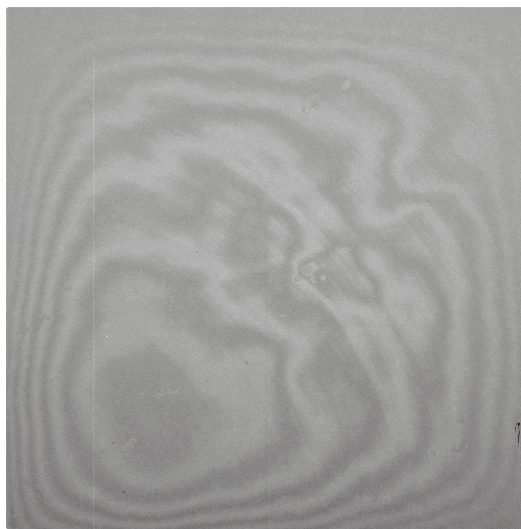
Any remaining variations between pixels are due to sensitivity variations and can be scaled/divided out of the data

Fringing

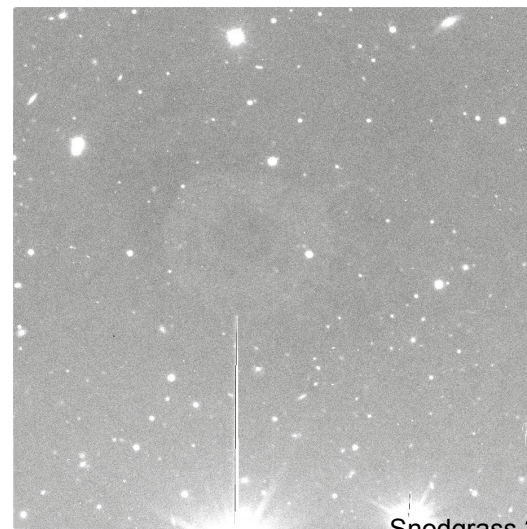
Interference of night sky lines (mostly in the redder optical and infrared bands, but sometimes also in narrowband filters) in the detector material can cause brighter and darker pattern in the sky background. Can be modeled and **subtracted** from the data during reduction.



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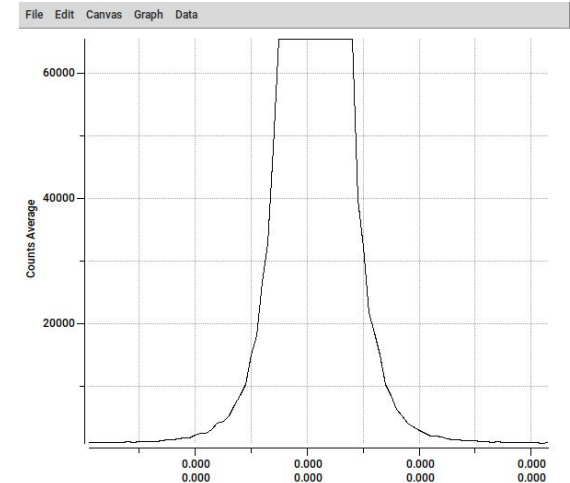
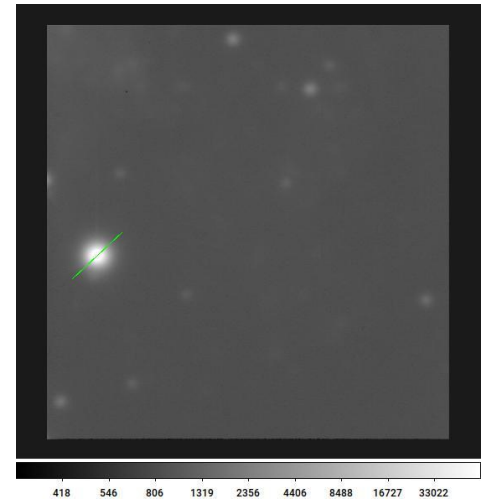


Saturation, full well depth & blooming

2 types of saturation: Numerical and physical

Numerical saturation is when an inappropriate gain leads to full pixels not using the entire dynamic range of the ADC → Bright stars show ~ flat tops at levels below 65535 counts

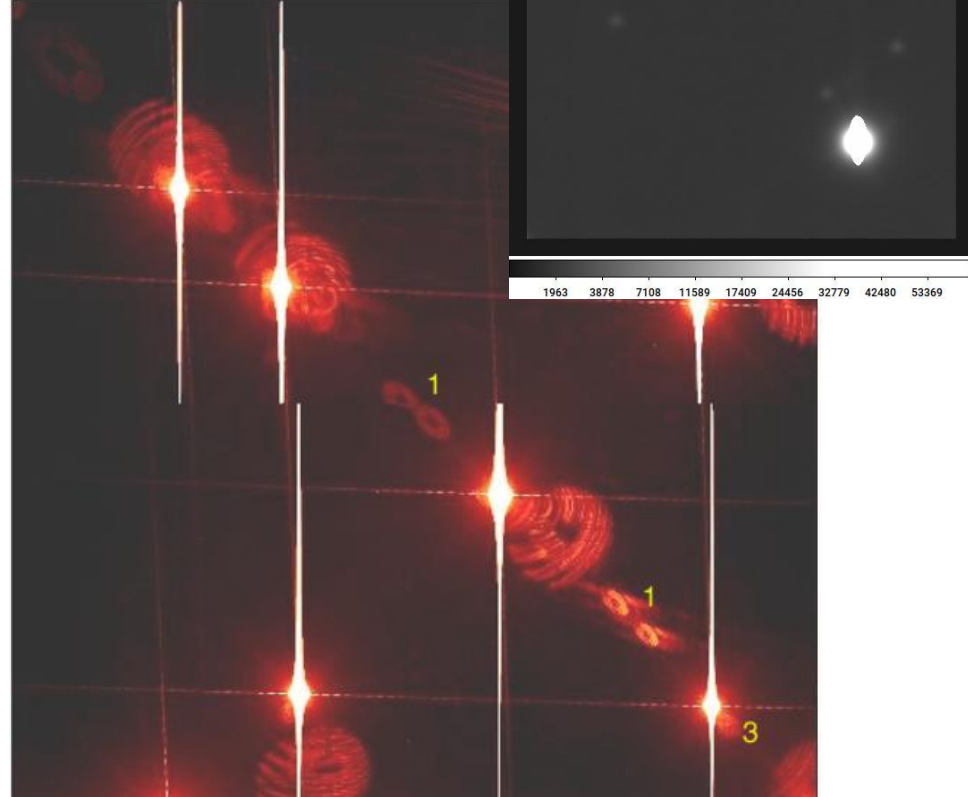
Physical saturation is when a source fills up the entire full well depth of a pixel. Adding any more light then can no longer be detected.



Saturation, full well depth & blooming

What happens when you overfill a pixel with charge?

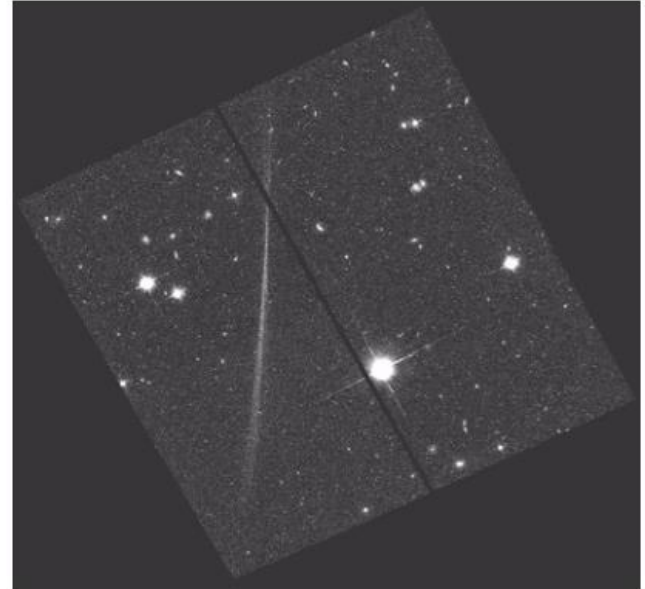
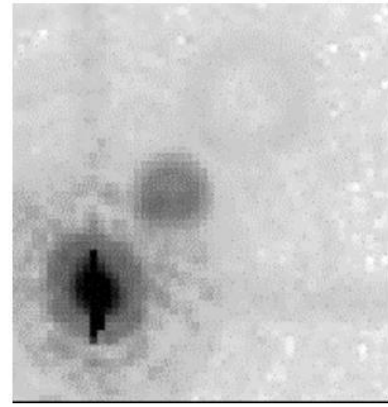
Charge typically can't move across columns due to channel gaps, but can leak out along columns and across rows.



Ghosts and glints

Ghosts are images caused by internal reflections of bright sources. Typically out of focus and offset from the actual source

Glints are also due to reflections of sources outside the field-of-view, typically due to bright stars and/or the moon.

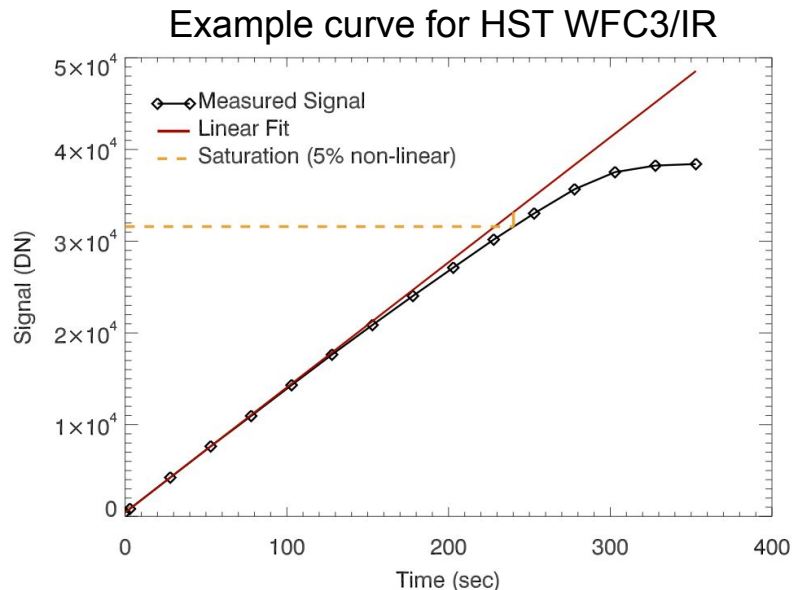


Detector Linearity

In an ideal world, doubling the input signal or exposure time would lead to a doubling of the measured output signal.

In reality this is only approximate and/or true only at lower intensities, with significant deviations at large intensities.

With sufficient data this can be measured, characterized, and then corrected for in the actual science data.

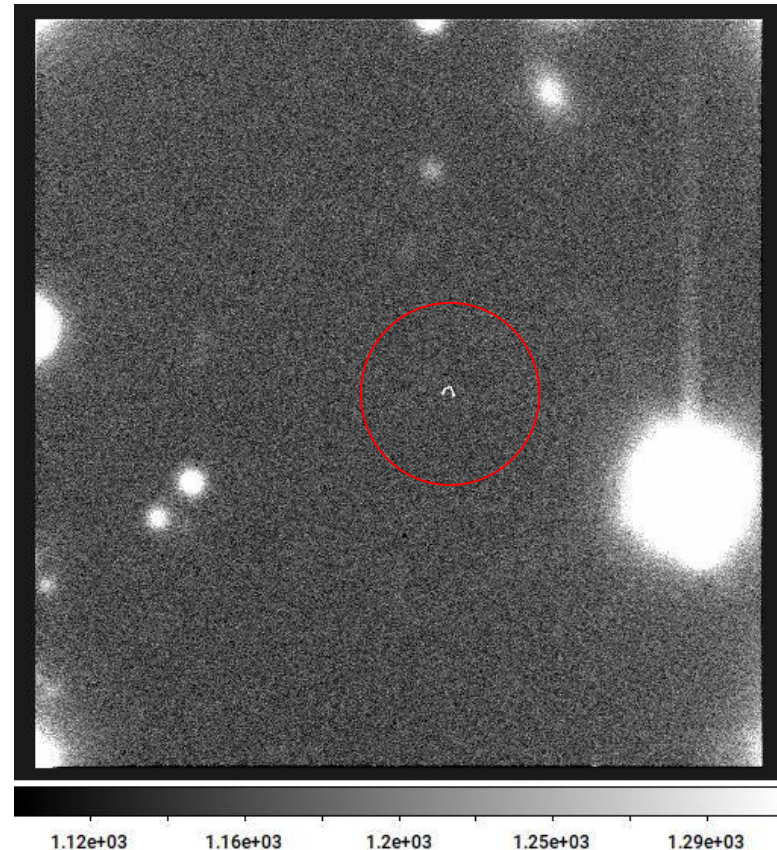


Transient effects: Cosmic Rays

In addition to light and dark current, CCDs are also particle detectors → Cosmics

Can be identified from their sharp edges.
No way to remove them from the image, but can either mask them out or fill them in with the average intensity of their surroundings.

Problematic once pixels become too large or undersampled and stars start to look like cosmics (e.g. HST WFPC2).



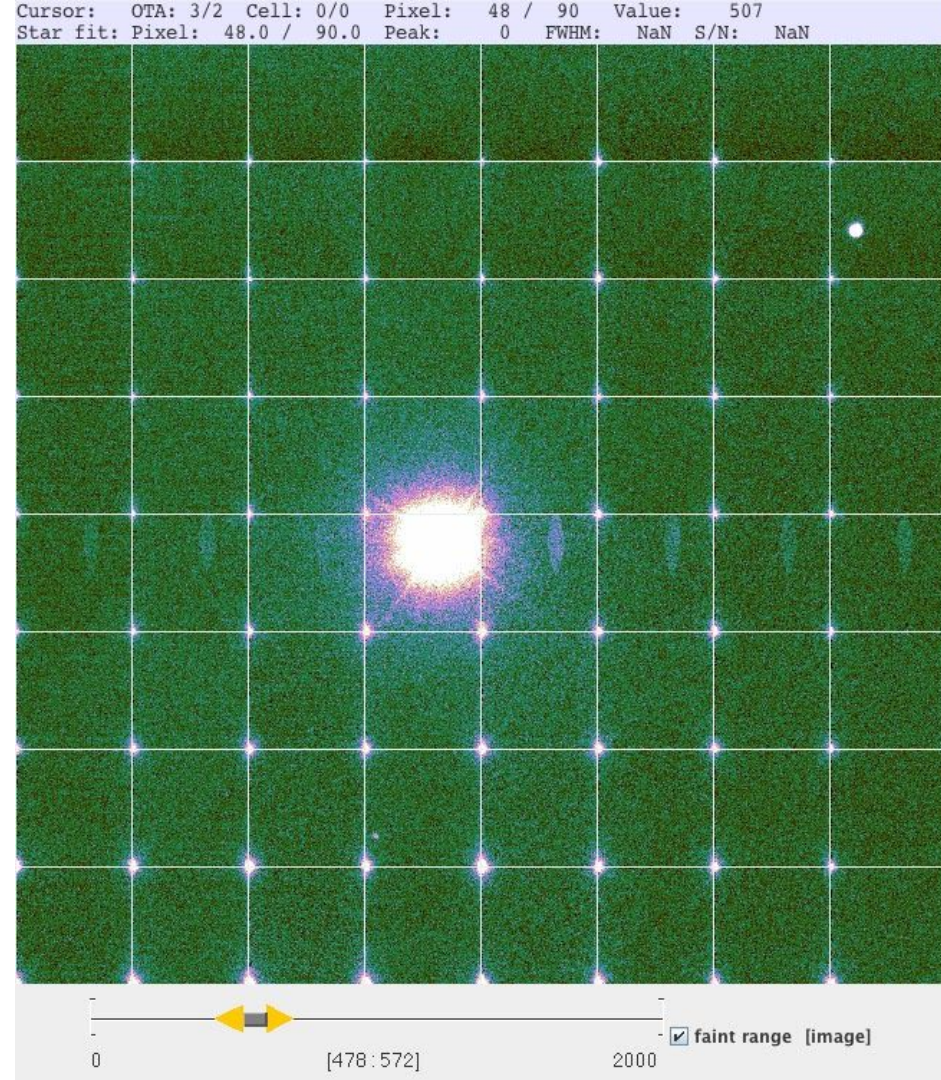


2003 Oct 31 20:30:12

Transient Effects: Cross-talk

Cross-talk happens when a strong signal read out in one ADC affects the signal registered in another ADC read out simultaneously.

Can be corrected in software after data is taken if cross-talk coefficients are known.

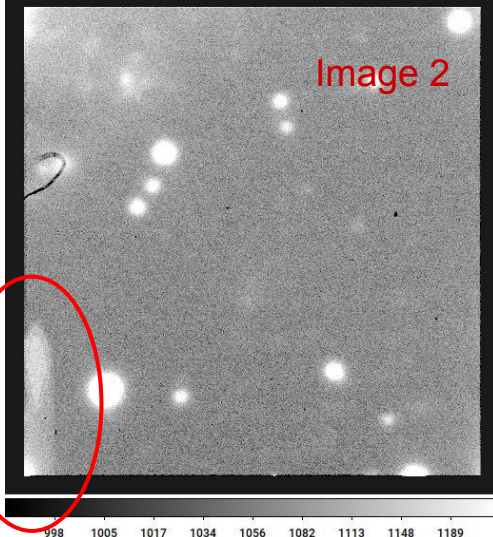
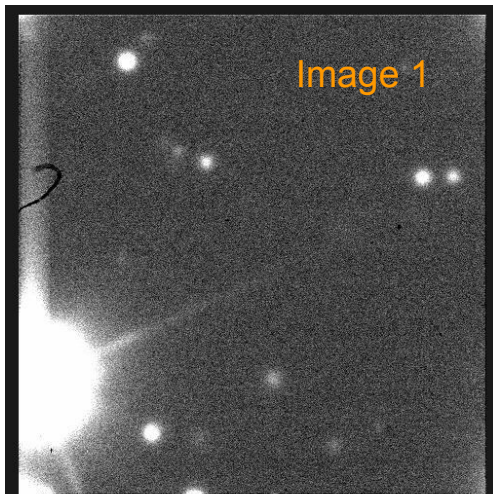


Transient effects: Persistence

Bright sources – especially when saturated – can leave some electron-hole pairs behind during the readout process. These can often be picked up during subsequent reads, generating ghost images.

Best to avoid them if possible (choose exposure times appropriately), or mask affected pixels to exclude it from the analysis

ghost image, from bright source in
Image 1, seen in Image 2

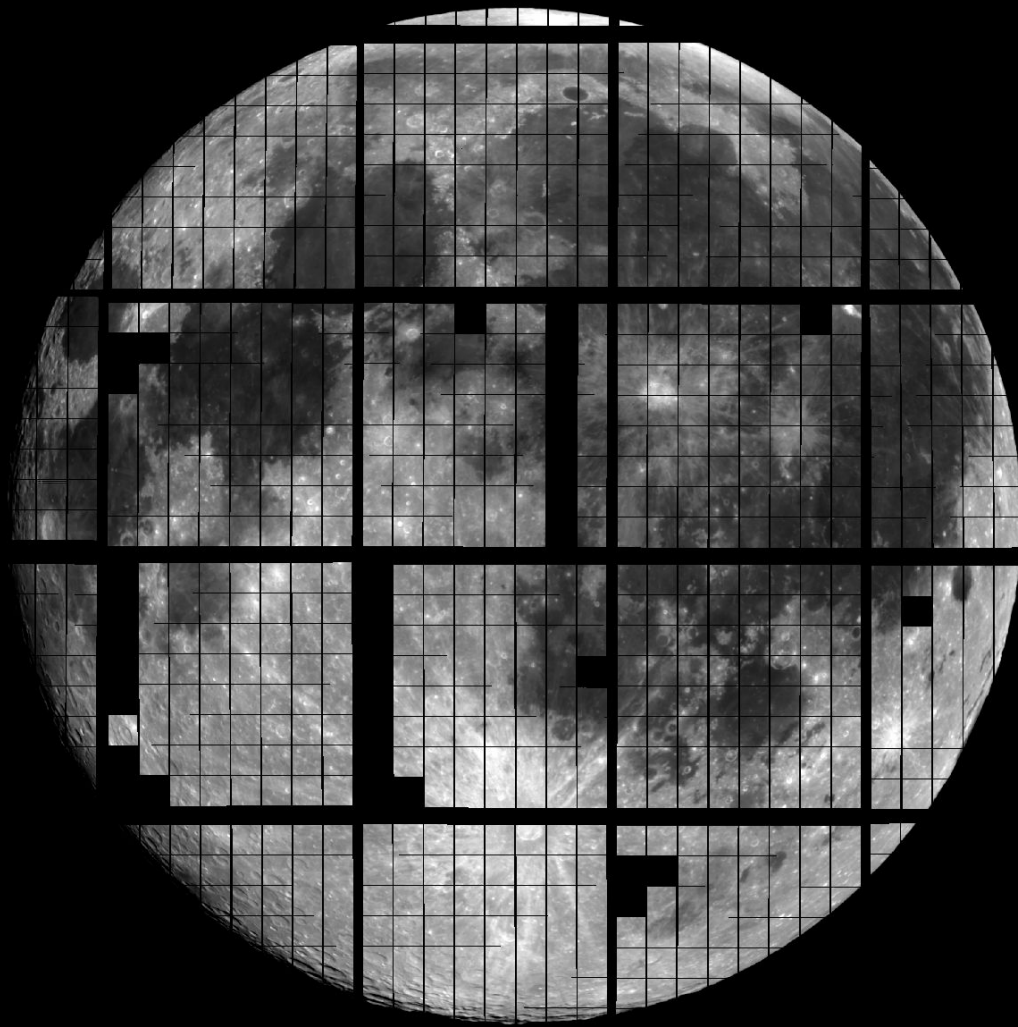


Basic data reduction

Data reduction removes instrumental artifacts to reconstruct/isolate the true scientific signal.

Order of processing steps is important! Generally follow the exposure and read process in reverse:

1. subtract bias (and dark)
2. subtract crosstalk (not needed for our data)
3. divide by flatfield
4. subtract additional contaminants if possible (fringing, glints, ghosts) – not req'd
5. subtract sky background
6. apply any flux corrections and calibrations
7. if available/required: combine multiple images



Noise in Astronomical Data

Read-noise: Sampling measurements → Gaussian/Normal distribution

Shot-/Photon-noise: Counting discrete events (i.e. photon detections)

→ Poisson distribution:

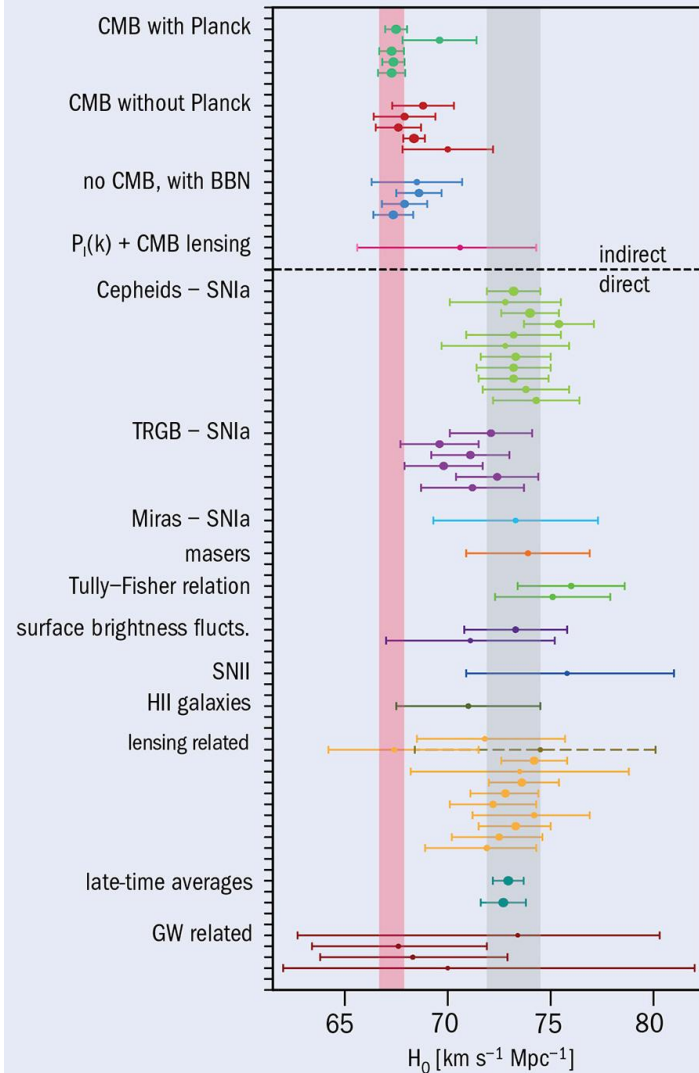
Standard deviation = square root of measured value

Why do we care: Measurements without uncertainties (or error estimates) are meaningless in science. Examples:

- Comparing results from different groups/studies
- Applying for telescope time

Example: Hubble tension

Different techniques to measure the local Hubble constant yield different values that are seemingly at odds with each other.



Signal and Noise: the CCD equation

What sources contribute to noise in images:

1) Signal due to arriving photons:

From the actual science source $S_{\star}$ [in electrons]: $\sigma_{\star} = \sqrt{S_{\star}}$

There is also another source of signal: sky background S_{sky}

$$\sigma_{\text{sky}} = \sqrt{S_{\text{sky}}}$$

Signal and Noise: the CCD equation

We also register signal NOT due to arriving photons:

Dark Signal, $S_D = t \times d$ where the dark current d is typically expressed in units of e-/pixel/hour.

Since just like light signal we are counting “photons”, the noise is given by

$$\sigma_D = \sqrt{S_D} = \sqrt{t \times d}$$

And finally, we have our readnoise that is not associated with any signal: R

Signal and Noise: the CCD equation

After readout, we can not tell the source of the different components apart, all we measure is the combined signal and its associated combined noise:

$$N = \sqrt{\sigma_{\star}^2 + \sigma_{\text{sky}}^2 + \sigma_D^2 + R^2}$$

$$N = \sqrt{S_{\star} + S_{\text{sky}} + t \times d + R^2}$$

The S/N for this **single-pixel** measurement can then be expressed as:

$$\frac{S}{N} = \frac{S_{\star}}{\sqrt{S_{\star} + S_{\text{sky}} + t \times d + R^2}}$$

Signal and Noise: the CCD equation

In most cases the astronomical signal we are interested in is measured across many pixels n_{pix} . We can then expand the previous equation to

$$\frac{S}{N} = \frac{S_{\star}}{\sqrt{S_{\star} + n_{\text{pix}} \times (S_{\text{sky}} + t \times d + R^2)}}$$

This equation is informally referred to as the **CCD equation**.

Note that for larger number of pixels the read-noise becomes more and more critical, since it enters linearly while the other terms only enter as square root of their signals.

$$\frac{S}{N} = \frac{S_{\star}}{\sqrt{S_{\star} + n_{\text{pix}} \times (S_{\text{sky}} + t \times d + R^2)}}$$

However, for the previous CCD equation to hold we would have to know the sky background level with absolute precision.

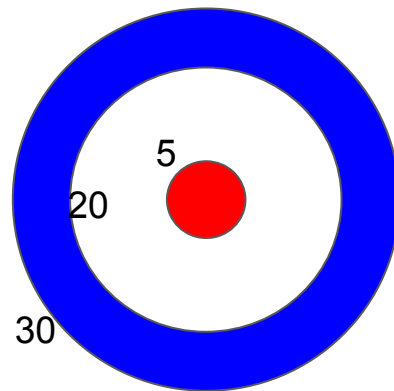
If we also account for uncertainties in the determination of the sky background level – averaged over n_{sky} pixels, we need to modify the equation as follows:

$$\frac{S}{N} = \frac{S_{\star}}{\sqrt{S_{\star} + n_{\text{pix}} \times \left(1 + \frac{n_{\text{pix}}}{n_{\text{sky}}}\right) \times (S_{\text{sky}} + t \times d + R^2)}}$$

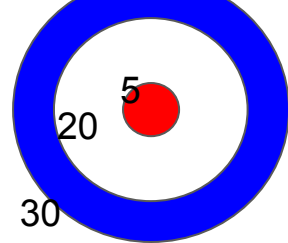
The last correction becomes small when n_{sky} is much larger than n_{pix} , so we should try to average over LOTS of sky pixels to estimate a reliable sky background level.

Example calculations

Problem: Compute the S/N obtained for an astronomical source in a CCD image with an exposure time $t = 300$ sec. The CCD has a gain of $2 \text{ e}^- / \text{ADU}$, a read-noise of 6 e^- , and a dark rate of $10 \text{ e}^- / \text{pix/hr}$. The sky background level is measured to be **604** ADU, by averaging the pixel values in an annulus of blank sky around the object of interest with an inner radius of 20 pixels and outer radius of 30 pixels. The signal due to the source, measured in an aperture of radius 5 pixels, is found to be $S_{\star+\text{sky}} - S_{\text{sky}} = \textbf{6487}$ ADU.



Example calculations



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$$\frac{n_{\text{pix}}}{n_{\text{sky}}} = \frac{\pi \cdot 5^2}{\pi \cdot (30^2 - 20^2)} = 0.05$$

$$n_{\text{pix}} \cdot \left(1 + \frac{n_{\text{pix}}}{n_{\text{sky}}}\right) \approx n_{\text{pix}} = \pi \cdot 5^2$$

$$d = 10 \text{ e}^-/\text{hr} = \frac{10}{3600} \text{ e}^-/\text{sec}$$

$$\frac{S}{N} = \frac{6487 \cdot 2}{\sqrt{(6487 \cdot 2) + \pi \cdot 5^2 \cdot \left(604 \cdot 2 + 300 \cdot \frac{10}{3600} + 6^2\right)}} \sim 39$$

This is quite a bit smaller than $S_{\star}/\sqrt{S_{\star}} \sim 114$, demonstrating that is important to account for all uncertainties when estimating S/N.